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SUBSTATION INSULATORS

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

This Engineering Standard establishes criteria in the selection and application of post type insulators, apparatus bushings, hollow insulators and wall bushings to be installed in indoor or outdoor substations and switching stations in SEC system.

2.0 OPERATIONAL REQUIREMENTS

2.1 Site conditions

SEC power system operates in an onerous pollution environment. The insulators shall be suitable for operation in site conditions specified in TES-P-119.02.

2.2 Contamination Levels

The insulators covered by this standard shall be applied in an environment with contamination levels indicated in TES-P-119.02, without flashover.

2.3 Creepage Distances

2.3.1 Outdoor Insulators shall have a minimum creepage distance as specified in 01-TMSS-01. Higher creepage distance if required, shall be specified in the project Scope of Work/Technical Specifications (SOW/TS).

2.3.2 Indoor insulators shall have a minimum creepage distance as specified in 01-TMSS-01.

2.4 Mounting Arrangement

2.4.1 Insulator mounting shall be arranged either upright, horizontal or underhung as specified in the SOW/TS. For outdoor insulators moisture from rain/fog or insulator washing shall not be trapped on the underside of the insulator skirts. Orientation of the insulator shall be in such a manner that the accumulation of air borne dust, sand, salt and industrial contaminants is minimized.

2.4.2 When the insulators are mounted in horizontal and underhung position, weight of the insulators, fittings, busbars and any other supported equipment shall be considered to determine the permissible load. Factor of safety given in this standard shall be met by the insulator in upright, horizontal and overhung positions.

2.5 Mechanical Characteristics

Insulators shall meet the mechanical strength requirement with factor of safety as required for each type. Calculations shall be submitted to prove the adequacy of insulator mechanical strength.

3.0 STATION POST INSULATORS

Station post insulators are used for the support of disconnect switch, tubular bus, air core reactors etc. Post insulators shall be wet-process porcelain type of brown color with a standard glaze. However for indoor application, other insulating materials with equal or better electrical and mechanical characteristics compared with wet-process porcelain can be proposed subject to SEC approval.

3.1 Stacking

Uniform diameter stacking type post insulators shall be used mainly for disconnect switches for 110kV, 115kV and 132kV. Single non-stacking type insulators shall be used for 69kV. For application other than disconnect switch, staking or non-stacking type insulators shall be used for 110kV, 115kV and 132kV. Post insulators above 132kV shall be stacking type for all applications.

3.2 Electrical Characteristics

The electrical characteristics of the outdoor and indoor post insulators shall be in accordance with TES-P-119.02.

3.3 Mechanical Characteristics

3.3.1 Normal working load (i.e. excluding short circuit forces) shall not exceed the following percentage of post insulator rated strength:

Cantilever	: 40%	Tension	: 50%
Compression	: 50%	Torsion	: 40%

3.3.2 Under short circuit conditions, the resultant of cantilever forces due to working load and short circuit load multiplied by their respective overloading factors shall not exceed the post insulator cantilever strength. Overloading factor for working load during short circuit shall be 2.5. Overloading factor for short circuit loads for post insulators with rigid bus connection shall be taken as 1.0 as specified in ANSI/IEEE 605, provided the following conditions are satisfied:

a. The natural frequency of the insulator together with the effective weight of the conductor span (f_i) is less than one-half of the short-circuit current-force frequency , i.e:

$$f_i < \frac{120}{2} \text{ Hz}$$

Where f_i = Natural frequency of the insulator with effective weight of conductor span in Hz.

b. Natural frequency of the insulator/mounting structure combination (f_{s1} and f_{s2}) and natural frequency of the conductor span (f_b) differ by a factor of atleast 2, i.e. :

$$\frac{f_{s1}}{f_b} < \frac{1}{2} \text{ or } \frac{f_{s1}}{f_b} > 2$$

$$\frac{f_{s2}}{f_b} < \frac{1}{2} \text{ or } \frac{f_{s2}}{f_b} > 2$$

Where

f_{s1} = first natural frequency of insulator/mounting structure combination in Hz

f_{s2} = second natural frequency of insulator/mounting structure combination in Hz

f_b = natural frequency of conductor span in Hz

If either of these conditions is not satisfied, a dynamic study shall be made to determine an appropriate overload factor or a conservative factor of 2.5 shall be assumed.

4.0 APPARATUS BUSHINGS

Apparatus bushing are for indoor and outdoor application and they are of Oil to Air, Oil to Oil, Oil to SF_6 and SF_6 to Air type.

4.1 Outdoor Bushings

Outdoor bushings shall be wet-process porcelain type of brown color with a standard glaze. They shall have a creepage distance as specified in TES-P-119.02. Higher creepage distance if required, shall be specified in SOW/TS. Outdoor bushings are of Oil to Air or SF_6 to Air type. Oil to Air type bushing rated above 69kV shall be oil impregnated paper bushing (OIP) type and for rated voltage 34.5kV and below it shall be normal porcelain type. SF_6 to Air bushings shall be resin impregnated paper (RIP) type.

4.1.1 Electrical Characteristics

Electrical characteristics of the outdoor bushings shall be in accordance with the insulation level specified in TES-P-119.02. Bushing rated current shall meet the application requirement with overloading as required. Rated thermal short time current (I_{th}) for transformer bushings shall be 25 times bushing rated current for 2 seconds per IEEE C57.109.

Thermal short time current for other bushings shall be 25 times rated current for one second. Dynamic withstand current rating of bushing other than transformer bushing shall be 2.6 times equipment short circuit current and for for transformer bushing dynamic withstand current shall be per IEC 60076-5 or equivalent standards.

4.1.2 Mechanical Characteristics

The maximum permissible mechanical loading applied to bushing terminal shall not exceed the value as specified in Table 11-1. Terminal loading during normal condition shall not exceed 50% of the tested value. The cantilever loading applied to the bushing terminal as a result of maximum permissible mechanical loading plus loading due to short circuit forces shall not exceed bushing tested value with a factor of safety of 1.5.

4.1.3 Termination

Apparatus bushing shall be terminated with flexible conductors only. In exceptional cases tubular busbar can be terminated to bushings with prior approval from SEC. Apparatus bushing with tubular connection shall meet the requirements of clause 3.3.

4.2 Cable Box Bushings

Cable box bushing shall be oil to air or oil to oil type. Oil to oil type bushings shall normally be provided for cable box rated above 69kV and it shall be RIP type. For rated voltage 34.5kV and below cable box bushing other than station service transformer, shall be wet processed porcelain type of brown color. For station service transformer, primary voltage bushing shall be either wet proceesed porcelain or dead-break separable insulated connector system type per IEEE 386 as specified in SOW/TS.

4.2.1 Electrical Characteristics

Electrical characteristics of the outdoor bushings shall be in accordance with the insulation level specified in TES-P-119.02. Bushing rated current shall meet the application with overloading as required. Thermal short time current rating and duration shall be per clause 4.1.1.

4.2.2 Mechanical Characteristics

The maximum permissible mechanical loading applied to bushing terminal shall not exceed the value as specified in Table 11-1. Other mechanical loading requirements of bushing shall be per clause 4.1.2.

4.3 Indoor Bushings

Apparatus bushings for indoor substation equipment shall be wet-process porcelain type of brown color with standard glaze. Indoor bushings shall be used for maximum voltage up to 69kV. Proposals for other insulating material with equal or better electrical and mechanical characteristics compared wet-process porcelain shall be subject to SEC approval.

4.3.1 Electrical Characteristics

Electrical characteristics of indoor bushing shall be in accordance with TES-P-119.02. Bushing rated current shall meet the application requirement with overloading as required. Thermal short time current rating and duration shall be per clause 4.1.1.

4.3.2 Mechanical Characteristics

The maximum permissible mechanical loading applied to a bushing terminal connection shall be as specified in clause 4.1.2.

Table 11-1 : Maximum Permissible Operating Load for Apparatus Bushing

Rated Voltage (KV_{rms})	Rated Continuous Current (A)			
	800	1000 to 1600	2000 to 2500	≥3150
	**Cantilever Operating load (N)*			
34.5	500	625	1000	1575
72.5	500	625	1000	2000
110 to 132	625	800	1250	2000
230	625	800	1250	2000
380	1250	1250	1575	2500

Note : * When the bushing is installed at an angle not exceeding 30º from vertical

** When the cantilever operating load exceeds the value given in the table, heavy load bushing per IEC 60137 shall be selected.

4.3.3 Termination

Apparatus bushing shall be terminated with flexible conductors only. In exceptional cases tubular busbar can be terminated to the bushings with prior approval from SEC. Apparatus bushing with tubular connection shall meet the requirements of clause 3.3.

5.0 HOLLOW INSULATORS

Hollow insulators are intended for use in circuit breakers, instruments transformers, surge arresters etc. Hollow insulators used in application such as circuit breakers etc., are with permanent gas pressure.

5.1 Electrical Characteristics

Dielectric withstand capability of equipment hollow insulators other than surge arrester shall be per TES-P-119.02. Surge arrester housing dielectric withstand capability shall be per IEC/IEEE standards. Short circuit withstand capability of surge arrester housing shall be higher than the system short circuit level in order to ensure the housing is not shattered during fault. All other characteristics of the hollow insulators including surge arrester housing shall be per TES-P-119.02.

5.2 Mechanical Characteristics

Factors such as mass, internal pressure, terminal loads, wind loads, operating loads and short circuit loads contribute to the bending stress of hollow insulators. The following sources shall be used for calculation of relevant loads:

- terminal load :	Clause 6.101.6 of IEC 62271-100 or equivalent standards
- wind loads :	Clause 6.101.6.1 of IEC 62271-100 and 2.1.2 of IEC 60694 or equivalent standards
- operating loads:	Depending on the design of equipment
- short circuit loads :	Determined from the rated short circuit level of the equipment and to be calculated per Section 2 of IEC 60865-1 or equivalent standards.

The load combinations to be considered for the mechanical characteristics of the hollow insulators shall be per Table 11-2. The factor of safety for routinely expected loads such as wind etc., and rarely occurring extreme loads such as short circuit force shall be per Table 11-2.

Table 11-2 : Hollow insulators loading combinations and weighting factors

Loads	Percentage of stress from routinely expected loads (%)	Percentage of stress from short circuit load (rarely occuring extreme load) (%)
Design pressure	100	100
Mass	100	100
Rated terminal load	100	50
Wind pressure	30	100
Short-circuit load	Not Applicable	100
Safety Factor	2.1	1.5

5.3 Termination

Hollow insulators shall be terminated with flexible conductor only. In exceptional cases tubular busbar can be terminated to hollow insulators with prior approval from SEC. Hollow insulator with tubular connection shall meet the requirement of clause 3.3.

6.0 WALL BUSHINGS

Wall bushings for indoor and outdoor use shall be wet-process porcelain type of brown color with a standard glaze. They shall have a creepage distance as specified in TES-P-119.02.

6.1 Electrical Characteristics

Electrical characteristics of the wall bushings shall be in accordance with the insulation levels specified in TES-P-119.02. Bushing rated current shall meet the application requirement. Thermal short time current for the bushings shall be associated with transformer circuits shall be 25 times rated current for two seconds and for other applications it shall be 25 times rated current for one second.

6.2 Mechanical Loading

The maximum permissible mechanical loading applied to a bushing terminal connection shall be as specified in clause 4.1.2 of this standard.

7.0 BIBLIOGRAPHY

1. ANSI C.2, "National Electrical Safety Code", 2002.

2. ANSI/NEMA C29.1, "Test Methods for Electrical Power Insulators",1988 (R2002).

3. ANSI/NEMA C29.9, "Wet-Process Porcelain Insulators (Apparatus, Post Type)", 1983 (R2002).

4. ANSI/IEEE C57.19.00, "General Requirements and Test Procedure for Outdoor Power Apparatus Bushings", 2004.

5. ANSI/IEEE C57.19.01, "Standard Performance Characteristics and Dimensions for Outdoor Apparatus Bushings", 2000 (R 2005).

6. ANSI/IEEE C57.19.100, “Guide for Application of Power Apparatus Bushing”, 1995 (R2003).

7. IEC 60137, "Insulated Bushings for Alternating Voltages above 1000 V", 2003.

8. IEC 60168, "Tests on Indoor and Outdoor Post Insulators of Ceramic Material or Glass for System with Nominal Voltages Greater than 1000 V", 2001.

9. IEC 60273, "Characteristics of Indoor and Outdoor Post Insulators for System with Nominal Voltage greater than 1000V”, 1990.

10. IEC 60865-1, "Short-circuit Currents - Calculation of Effects, Part 1: Definitions and Calculation Methods". Second Edition 1993 and Corrigendum March 1995.

11. IEC 60865-2, "Short-circuit Currents - Calculation of Effects, Part 2: Examples of Calculation". 1994

12. IEEE 605, "Guide for Design of Substation Rigid - Bus Structures," 1998.

13. IEC 62155, “Hollow Pressurized and Unpressurized Ceramic and Glass Insulators for Use in Electrical Equipment with rated Voltages greater than 1000V”, 2003.